A Dirichlet-Convolution Identity for OEIS A001222

An elementary proof of the two-parameter identity

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Abstract

We give a self-contained proof of the two-parameter divisor-sum identity recorded as a conjecture in OEIS A001222. The statement is valid for arbitrary complex parameters, including zero bases with the convention $0^0 = 1$. The argument reduces a Dirichlet convolution of multiplicative functions to a finite difference-of-powers identity at prime powers. An existing formal proof has been reported elsewhere; no claim of novelty or priority is made.

Keywords: prime factors, Dirichlet convolution, multiplicative functions, OEIS A001222.

1 Statement

Let $\Omega(n)$ denote the number of prime factors of n, counted with multiplicity (OEIS A001222), and let $\omega(n)$ denote the number of distinct prime factors (OEIS A001221). Then OEIS A046660 is $\Omega(n)-\omega(n)$. In all expressions below, define $0^0 = 1$.

Theorem 1. *For every $x, y \in \mathbb{C}$ and every positive integer n,*

$$\boxed{(x+y)^{\Omega(n)} = \sum_{d\mid n} x^{\Omega(d)} (x+y)^{\Omega(n/d)-\omega(n/d)} y^{\omega(n/d)}.} \tag{1}$$

2 Proof

Define the arithmetic functions

$$F(n) = (x+y)^{\Omega(n)}, \qquad G(n) = x^{\Omega(n)}, \qquad H(n) = (x+y)^{\Omega(n)-\omega(n)} y^{\omega(n)}.$$

Both Ω and ω are additive on coprime arguments. Therefore F, G, H are multiplicative functions, including the normalization $F(1) = G(1) = H(1) = 1$ (also when a base is zero). The Dirichlet convolution

$$(G * H)(n) = \sum_{d\mid n} G(d) H(n/d)$$

is multiplicative. Consequently it suffices to show $F(p^e) = (G*H)(p^e)$ for each prime p and integer $e \geq 0$.

For $e = 0$ both sides equal 1. For $e \geq 1$, the prime-power values are

$$G(p^j) = x^j, \qquad H(1) = 1, \qquad H(p^j) = y(x+y)^{j-1} \quad (j \geq 1).$$

It follows that

$$(G*H)(p^e) = \sum_{j=0}^{e} G(p^j) H(p^{e-j}) \tag{2}$$

$$= x^e + y \sum_{j=0}^{e-1} x^j (x+y)^{e-1-j} \tag{3}$$

$$= x^e + \big((x+y) - x\big) \sum_{j=0}^{e-1} x^j (x+y)^{e-1-j} \tag{4}$$

$$= x^e + \big((x+y)^e - x^e\big) = (x+y)^e = F(p^e). \tag{5}$$

The penultimate equality is the division-free difference-of-powers identity, so no nonzero assumption on x, y or $x+y$ is needed. Equality of multiplicative functions at all prime powers proves (1) for every $n \geq 1$. □

3 Scope and attribution

The OEIS entry labels this full complex-parameter identity a conjecture and cites an earlier special case. A separate formal proof of the general identity is already documented [4]. The present note is an elementary independent presentation of the same mathematical statement; it neither establishes publication priority nor claims to be the first proof. Mathematical steps were checked during preparation, but this PDF has not undergone independent peer review.

AI authorship and verification disclosure

Primary author (AI): ChatGPT (OpenAI). The exposition, calculations and preliminary checks were prepared using AI. Accuracy and originality should be evaluated independently before treating the note as a certified research contribution.

References

[1] OEIS Foundation, *A001222: Number of prime divisors of n counted with multiplicity*, t https://oeis.org/A001222 (consulted 11 November 2026).

[2] OEIS Foundation, *A001221: Number of distinct primes dividing n*, t https://oeis.org/A001221.

[3] OEIS Foundation, *A046660: Excess of n*, t https://oeis.org/A046660.

[4] trureturing, *Schulte's Ω/ω binomial convolution*, trureturing.org (research/oeis-a001222-schulte-omega-binomial-convolution/) (consulted 11 November 2026).